

CONFERENCE PRESENTATION

Evaluation of Probabilistic Prediction Performance in Water Quality Classification Using the Natural Gradient Boosting (NGBoost) Algorithm

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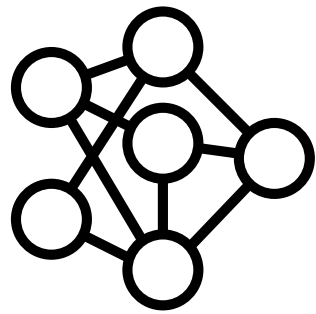
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Background

The quality of drinking water is very important for global public health [1]. However, there are several gaps, namely:

- The physicochemical parameters of water are multidimensional and often unstable [2].
- Deterministic ML models struggle to properly gauge prediction uncertainty [5].
- The transparency of ensemble models is still limited because of their black-box nature [6].

Research Question



Bagaimana performa NGBoost dibandingkan model ensemble deterministik pada klasifikasi kelayakan air?



Bagaimana perbandingan tingkat kalibrasi probabilitas antara NGBoost, XGBoost, dan Random Forest?



Bagaimana karakteristik ketidakpastian prediksi di sekitar ambang batas keputusan?

Metodologi & Data

Dataset & Distribusi

- The Water Potability Dataset contains 3,276 samples and 9 physicochemical parameters.
- Class Distribution: 60.99% Not Potable, 39.01% Potable.

Preprocessing

- Imputing missing values using MICE (keeping the multivariate structure).
- Stratified data splitting (70:15:15).
- Feature standardization (StandardScaler) on the training data.
- Note: SMOTE-ENN technique was evaluated but discarded because it significantly reduced model performance due to the drastic reduction in training data.

Configuration Model

Main Model (NGBoost)

- Using Bernoulli distribution parameterization.
- Objective function: Negative Log-Likelihood (NLL).
- Natural Gradient-based optimization on the Fisher Information Matrix.
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Baseline Models & Metrics

- XGBoost: A deterministic gradient boosting algorithm.
- Random Forest: A deterministic bagging algorithm.
- Evaluation Metrics:
 - Classification: Acc, Prec, Rec, F1, AUC

Probabilistic Calibration: NLL, ECE

Equivalent Classification Performance

Performa Klasifikasi Setara

- Using Bernoulli distribution parameterization.
- Objective function: Negative Log-Likelihood (NLL).
- Natural Gradient-based optimization on the Fisher Information Matrix.

Numerical Calibration Metrics

- Random Forest: Lowest ECE (0.042) and lowest NLL (0.618).
- NGBoost: ECE (0.069) and NLL (0.673).
- Findings: Surprisingly, NGBoost did not outperform the baseline model on numerical calibration metrics.

Analysis of the Zone of Uncertainty

Key Advantages of NGBoost: Quantification of uncertainty through explicit distribution parameterization.

Conclusion

- NGBoost dan model baseline memiliki performa klasifikasi yang setara secara statistik.
- Kerangka probabilistik natif NGBoost memungkinkan deteksi sampel dengan kepercayaan tinggi secara prinsipil dan terstruktur.
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Limitations of the Study

- NGBoost and the baseline model exhibit statistically equivalent classification performance.

NGBoost's native probabilistic framework enables the detection of high-confidence samples in a principled and structured manner.

Refernces

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